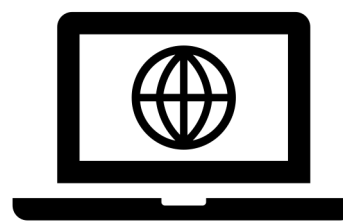


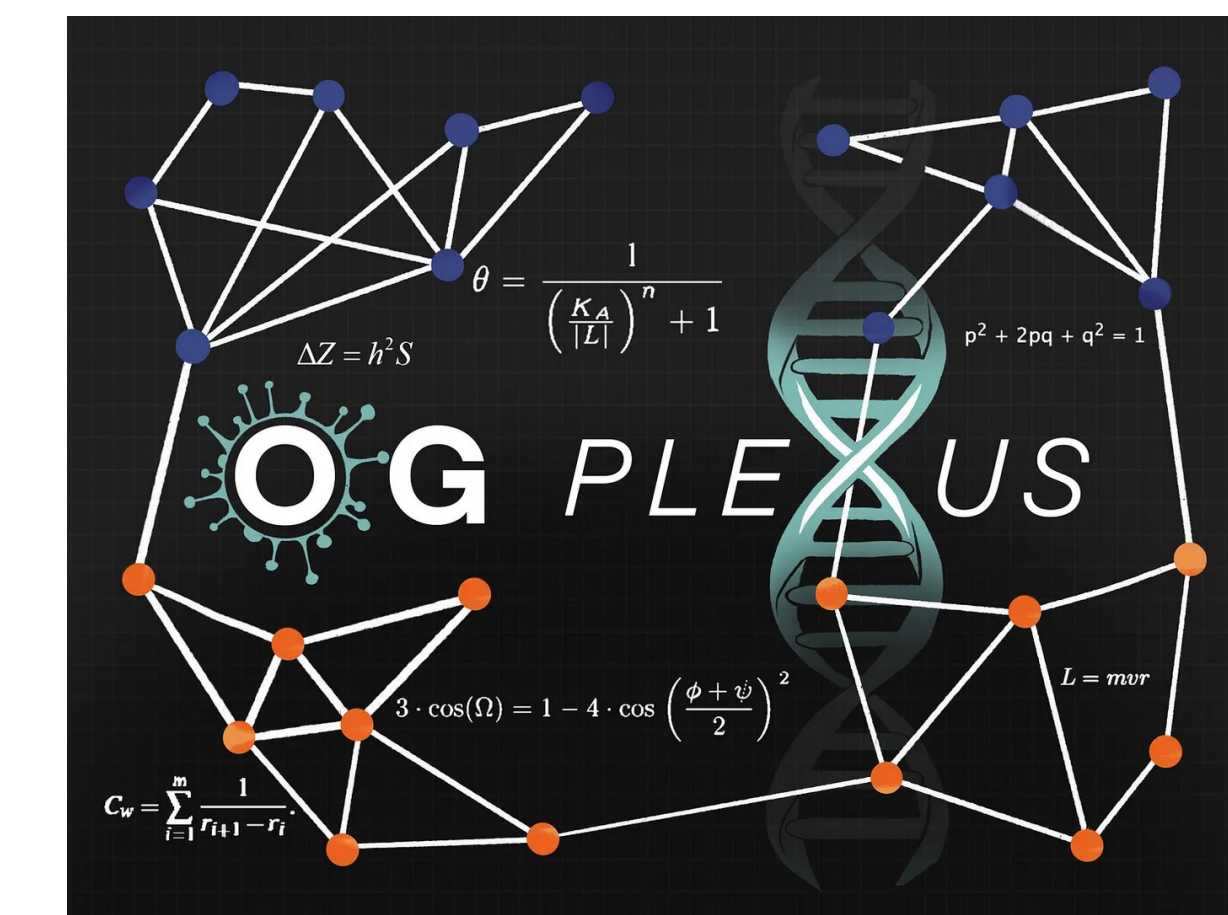
Deconstructing fitness seascapes across scales of granularity

Swathi Nachiar Manivannan¹, C. Brandon Ogbunugafor^{1,2,3}

¹Department of Ecology and Evolutionary Biology, Yale University; ²Public Health Modelling Unit, Yale School of Public Health; ³Santa Fe Institute

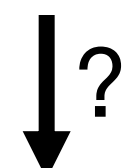


<https://swathi-nm.github.io>



Background

Environment



Genotype → Phenotype → Fitness

- Fitness landscapes: useful metaphor for visualizing genotype-fitness maps^{1,2}
- However, most fitness landscape studies tend to study evolution in constant, static environments³
- Fitness seascapes**: alternative metaphor that considers how **environmental context** shapes molecular evolution.^{3,4}
- Here, we ask the question: **How do changing environmental contexts influence features of seascapes across scales of granularity?**

Framework

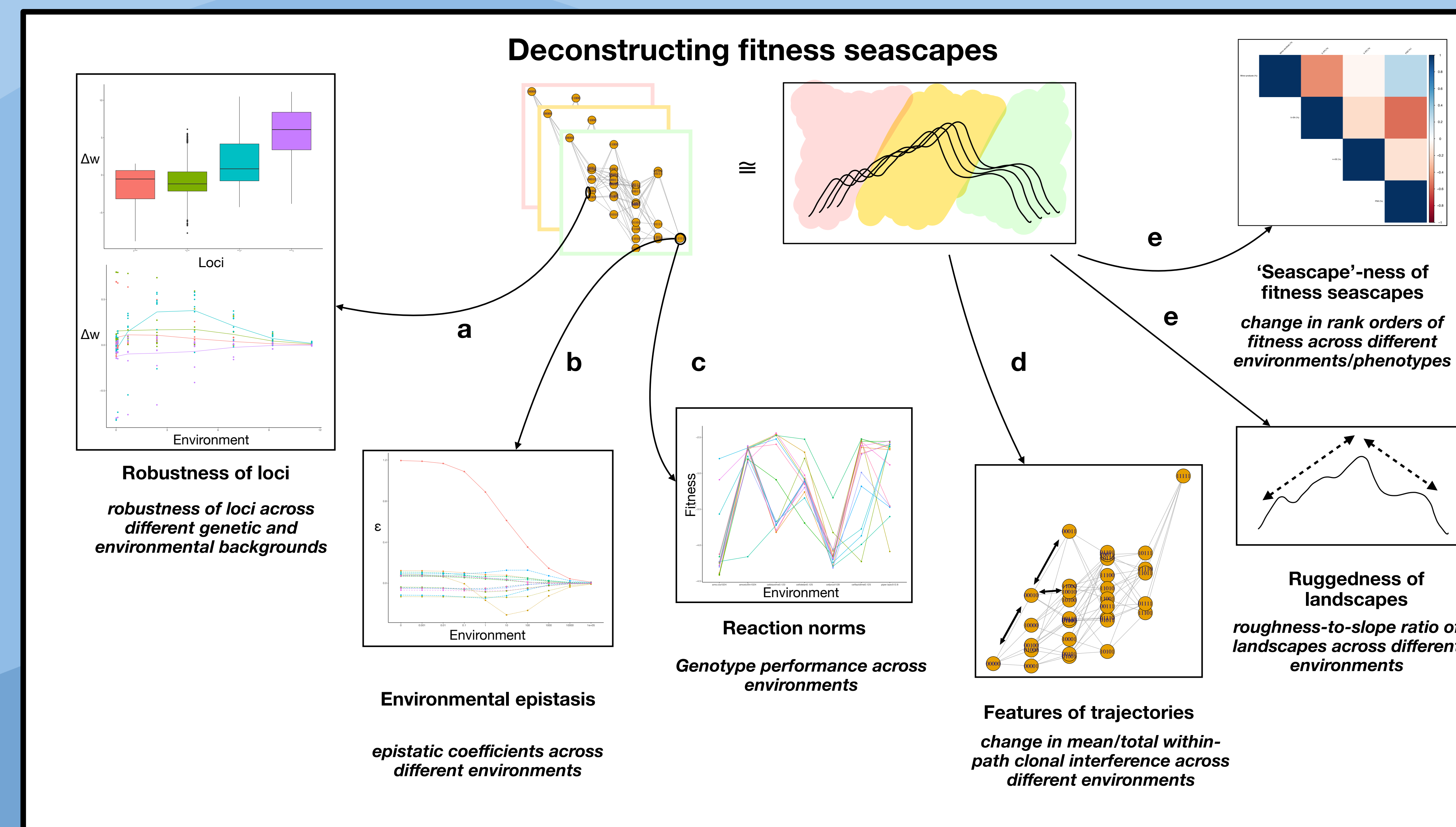


Fig 1. Framework for deconstructing seascapes into loci, locus interactions, alleles, trajectories and landscapes.

Conclusions

- We propose a quantitative framework to **disentangle** how **context-dependence and environmental dynamism** influences molecular evolution.
- We demonstrate this framework across a range of biological systems, and an initial comparative analysis suggests that the **allele** and **trajectory** scales are most **environmentally sensitive**.
- Understanding how and which scales of the fitness seascape alter can be instrumental for addressing questions such as **controlling microbial evolution**⁵ and/or **bioengineering applications**⁶.

Overview of study

- 12 empirical fitness seascape datasets that were **combinatorially complete** and ranged in size from **2³–2⁶ genotypes**.
- Deconstructed the seascapes into:
 - Loci** – genetic/environmental robustness
 - Locus interactions** – environmental epistasis
 - Alleles** – reaction norms
 - Trajectories** – within-path clonal interference
 - Landscapes** – ruggedness and 'seascape'-ness

How do different scales vary across environments?

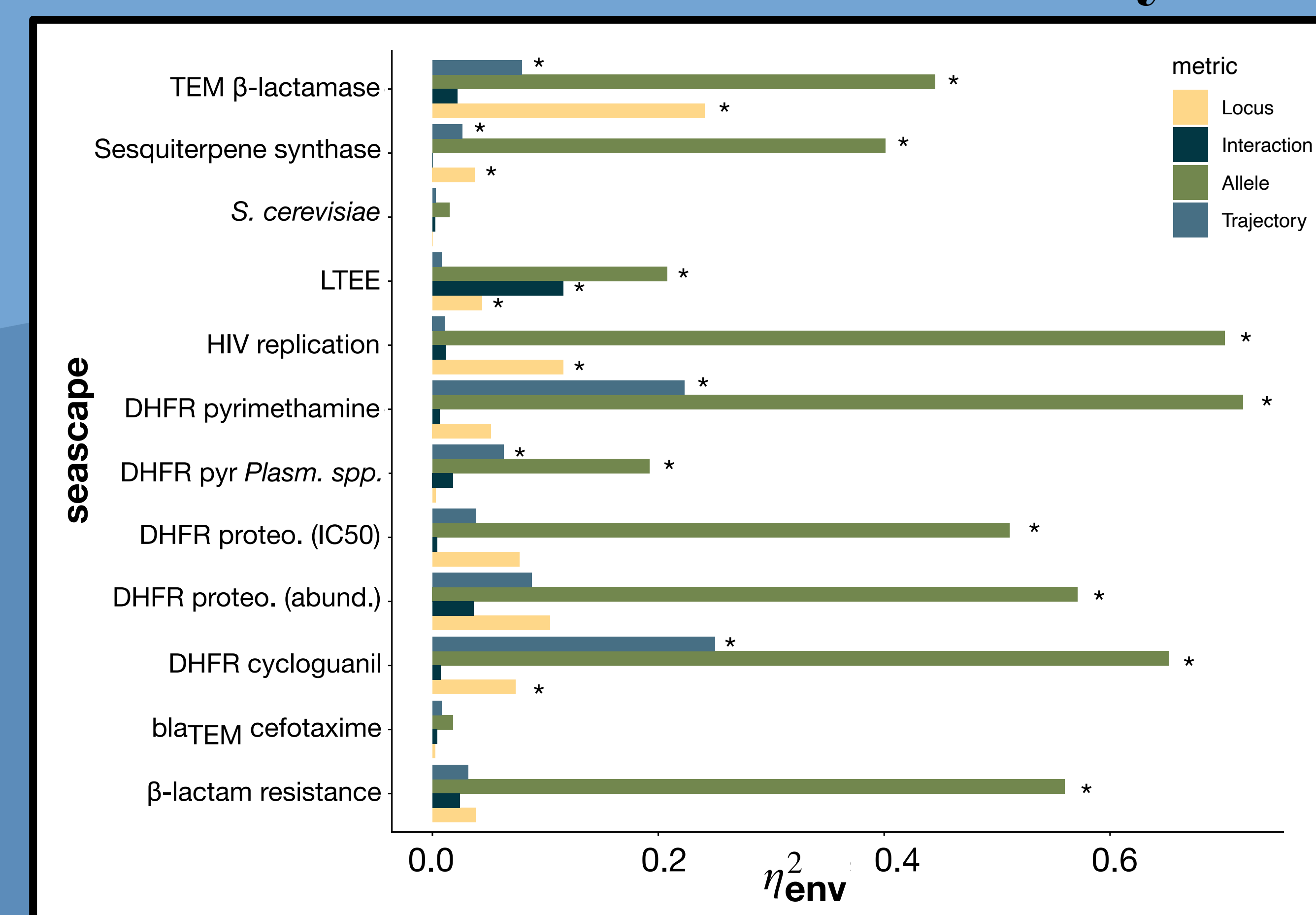


Fig 2. Environmental sensitivity (η^2_{env}) across scales for all seascapes.

- η^2_{env} varies substantially across scales and systems.
- Environmental effect statistically significant in **allele scale** for 10/12 seascapes, followed by **trajectories** (5/12 seascapes) and **loci** (5/12 seascapes).
- Overall genotype performance, accessibility of adaptive trajectories highly environmentally sensitive.

References & Acknowledgements

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For more information:

